Numerical processing of financial data

Additional topics: no arbitrage conditions on option prices and implied volatilities

Stefano De Marco

Ecole Polytechnique

Double degree Data and Finance, Feb 20th, 2026

In this lecture

- *Shape constraints* on option prices and implied volatilities
- The shape constraints are induced by *no-arbitrage conditions*:
  - on a call price surface $\mathrm{Call}(T, K)$, or a put price surface $\mathrm{Put}(T, K)$
  - equivalently, on an implied volatility surface $\sigma(T, K)$.

The market

We consider a market containing, as usual,

- A tradable asset S (for example a stock), a constant interest rate r.
- Forward contracts, with their forward prices $F_t^T = S_t e^{r(T-t)}$.
- Call options and put options traded at time t for any possible maturity $T > t$ and any possible strike $K > 0$.

The market

We consider a market containing, as usual,

- A tradable asset S (for example a stock), a constant interest rate r.
- Forward contracts, with their forward prices $F_t^T = S_t e^{r(T-t)}$.
- Call options and put options traded at time t for any possible maturity $T > t$ and any possible strike $K > 0$.
  - This is of course an idealization (in practice, only a finite set (T_i, K_i) of maturities and strikes is available in the market at a given time).
  - We denote $\mathrm{Call}_t(T, K)$ the call price, and $\mathrm{Put}_t(T, K)$ the put price, observed at time t for maturity T and strike K.

Option moneyness

- Let us denote

$$M_t^T = \frac{K}{F_t^T}$$

  the *forward moneyness* of an option with maturity T and strike K, seen at time t.

- Terminology: the at-the-money (ATM) point corresponds to

$$K = F_t^T$$

  hence to

$$M_t^T = 1 \qquad \text{or} \qquad \log M_t^T = 0$$

Option moneyness

- Let us denote

$$M_t^T = \frac{K}{F_t^T}$$

  the *forward moneyness* of an option with maturity T and strike K, seen at time t.

- Terminology: the at-the-money (ATM) point corresponds to

$$K = F_t^T$$

  hence to

$$M_t^T = 1 \qquad \text{or} \qquad \log M_t^T = 0$$

Remark: if the moneyness is fixed at some given value $M_t^T = M$, then this means that the option strike changes according to the maturity:

$$T \mapsto K = M\, F_t^T = M\, S_t\, e^{r(T-t)}.$$

In other words, we are considering a different option for every different maturity.

No-arbitrage conditions on Call prices

Proposition (No-arbitrage conditions on the call price surface)

Let t (the current observation date) be fixed. If there is no arbitrage on the market at time t, then the call price surface $(T, K) \mapsto \mathrm{Call}_t(T, K)$ must satisfy the following conditions:

0) *Bounds:*

$$(S_t - e^{-r(T-t)}K)^+ \leq \mathrm{Call}_t(T, K) < S_t, \qquad \forall\, T > t,\ \forall K\ > 0.$$

No-arbitrage conditions on Call prices

Proposition (No-arbitrage conditions on the call price surface)

Let t (the current observation date) be fixed. If there is no arbitrage on the market at time t, then the call price surface $(T, K) \mapsto \mathrm{Call}_t(T, K)$ must satisfy the following conditions:

0) *Bounds:*

$$(S_t - e^{-r(T-t)}K)^+ \leq \mathrm{Call}_t(T, K) < S_t, \qquad \forall\, T > t,\ \forall K\ > 0.$$

i) *Convexity in strike: the function $K \mapsto \mathrm{Call}_t(T, K)$ is convex, for all T.*

No-arbitrage conditions on Call prices

Proposition (No-arbitrage conditions on the call price surface)

Let t (the current observation date) be fixed. If there is no arbitrage on the market at time t, then the call price surface $(T, K) \mapsto \mathrm{Call}_t(T, K)$ must satisfy the following conditions:

0) *Bounds:*

$$(S_t - e^{-r(T-t)}K)^+ \leq \mathrm{Call}_t(T, K) < S_t, \qquad \forall\, T > t,\ \forall K\ > 0.$$

i) *Convexity in strike: the function $K \mapsto \mathrm{Call}_t(T, K)$ is convex, for all T.*

ii) *Monotonicity with respect to maturity: call options with fixed moneyness are increasing in T, that is: for every fixed moyeness M, the function*

$$T \mapsto \mathrm{Call}_t\big(T, M\, F_t^T\big)$$

is increasing.

No-arbitrage conditions, now on Put prices

Proposition (No-arbitrage conditions on the put price surface)

Let t (the current observation date) be fixed. If there is no arbitrage on the market at time t, then the put price surface $(T, K) \mapsto \mathrm{Put}_t(T, K)$ satisfies the following conditions:

0) *Bounds:*

$$(e^{-r(T-t)}K - S_t)^+ \leq \mathrm{Put}_t(T, K) < e^{-r(T-t)}K, \qquad \forall\, T > t,\ \forall K > 0.$$

No-arbitrage conditions, now on Put prices

Proposition (No-arbitrage conditions on the put price surface)

Let t (the current observation date) be fixed. If there is no arbitrage on the market at time t, then the put price surface $(T, K) \mapsto \mathrm{Put}_t(T, K)$ satisfies the following conditions:

0) *Bounds:*

$$(e^{-r(T-t)}K - S_t)^+ \leq \mathrm{Put}_t(T, K) < e^{-r(T-t)}K, \qquad \forall\, T > t,\ \forall K > 0.$$

i) *Convexity in strike: the function $K \mapsto \mathrm{Put}_t(T, K)$ is convex, for all T.*

No-arbitrage conditions, now on Put prices

Proposition (No-arbitrage conditions on the put price surface)

Let t (the current observation date) be fixed. If there is no arbitrage on the market at time t, then the put price surface $(T, K) \mapsto \mathrm{Put}_t(T, K)$ satisfies the following conditions:

0) *Bounds:*

$$(e^{-r(T-t)}K - S_t)^+ \leq \mathrm{Put}_t(T, K) < e^{-r(T-t)}K, \qquad \forall\, T > t,\ \forall K\ > 0.$$

i) *Convexity in strike: the function $K \mapsto \mathrm{Put}_t(T, K)$ is convex, for all T.*

ii) *Monotonicity with respect to maturity: put options with fixed moneyness are increasing in T, that is: for every fixed moyeness M, the function*

$$T \mapsto \mathrm{Put}_t\big(T, M\, F_t^T\big)$$

is increasing.

Proof: follows from the previous proposition for call options, using put-call parity $\mathrm{Call}_t(T, K) - \mathrm{Put}_t(T, K) = S_t - e^{-r(T-t)}K$.